

AI-Capella CS147 Final Report

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1. Project Name & Value Proposition

Project Name: AI-Capella

Value Proposition:

AI-Capella is a live voice-AI platform that helps a-cappella singers build confidence and skill through adaptive real-time feedback, personalized coaching, and guided daily lessons. By offering consistent, accessible, high-quality vocal guidance at any time, AI-Capella empowers singers to improve their performance while feeling empowered.

Team Members & Roles

Max R.

Class of '26 • Symbolic Systems & CS Cotermin (AI)

Roles:

- Team full stack website developer
- Focus on needfinding, technical report writing, and summation
- Figma frontend and backend development
- Video director, editor, and actor
- Prototype tester (computer operator and notetaker)

Marcus L.

Class of '25 • CS HCI Cotermin

Roles:

- Focus on needfinding synthesis
- Prototype testing
- Developing interviewing skills

- Working on high-fidelity prototype development

Vaishnav G.

Class of '25 • CS HCI Coterm

Roles:

- High-fidelity prototype react-native developer
- Medium-fidelity prototype front-end style design
- Needfinding and interview organizer
- Designed team presentations

Bryant M.

Class of '26 • CS

Roles:

- Focus on video editing for Assignment 4
- Conducted user interviews
- Developed medium-fidelity prototype design and functionalities

3. Problem / Solution Overview

In-person vocal instruction, which is essential for a cappella singers to build confidence and technical skill, is often too infrequent and expensive. Existing alternative instruction methods fall short in several ways: feedback can be overly harsh or generic, lacks personalization, is delayed rather than immediate, and tends to make vocal practice feel like a chore rather than an enjoyable activity. AI-Capella addresses this gap by offering a 24/7 AI vocal coach that delivers real-time, adaptive feedback and encouragement through generative voice technology. This approach creates an accessible, personalized, and supportive learning space tailored to each singer's unique needs and mistakes (updated on a daily basis).

4. Needfinding

a. Interviews

Participants and Recruitment

Our team conducted two rounds of semi-structured interviews and a set of short experience-prototype sessions. Participants were recruited formally through impersonal-online and mutual networks and participated voluntarily without monetary compensation. All interviews were

conducted in-person and lasted approximately 30–45 minutes. One team member led the conversation while another focused on detailed note-taking and capturing key quotes. We maintained a conversational tone and asked open-ended questions about participants' relationships with music, their experiences with singing and criticism, and their perceptions of AI in creative spaces.

Round One Participants:

- **Music Teacher:** Shared a strong dislike of AI in music, expressed concern about discouragement caused by overly critical feedback, and warned against over-reliance on AI tools.
- **AI Educator and Singer:** Felt mild discomfort with AI in creative work, believed AI should be a last-resort option, and described discomfort with auditions and harsh criticism.
- **Drummer:** Reported embarrassment and lack of confidence around singing, and regret over never attempting it seriously.
- **Composer:** Felt conflicted about AI in creative practice, yet acknowledged its industry-standard status and shared experiences of nearly quitting singing due to critical instruction.

Round Two Participants:

- **Music Lecturer (Stanford):** Traditional musician with minimal technological focus.
- **Ryan K.:** Intermediate jazz pianist and beginner vocalist who lost confidence after receiving evasive feedback from his vocal teacher.
- **Daniel:** AI engineer interested in voice technology who finds current voice AI interactions less natural than text-based AI.
- **Fernando:** Non-musician curious about learning but deterred by high activation energy and logistical barriers.

Experience Prototypes:

- Gamified a-capella app tested with two college students.
- Artistic voice diary reflection session with a social media influencer participant.
- Criticism-and-compliments coaching session between an experienced singer and a novice.

These sessions allowed us to observe real behaviors surrounding musical instruction-based vulnerability, motivation, and feedback styles in musical practice.

b. Synthesis

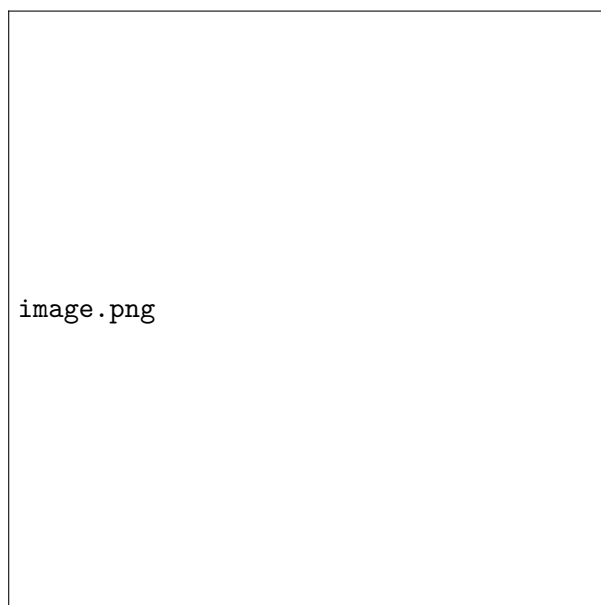


Figure 1: Empathy Map for Stanford Lecturer

To synthesize our findings, we created empathy maps for each participant, capturing what they said, thought, did, and felt regarding singing, feedback, and AI. This process revealed significant emotional patterns and allowed us to compare shared experiences across participants.

Key Insights:

- **Singing is emotionally intimate:** Participants consistently described feelings of vulnerability, embarrassment, and long-lasting impact from past criticism.
- **Confidence is the primary barrier:** Most participants were interested in singing but avoided it due to self-doubt and fear of judgment.
- **Voice AI for pedagogy and art feels emotionally flat:** Current voice technologies are perceived as transactional and unsupportive for creative growth.
- **High activation energy prevents starting:** Logistical and psychological barriers discourage individuals from pursuing vocal practice.
- **Balanced feedback is desired:** Participants sought supportive yet constructive feedback that motivates improvement.

Insights from our experience prototypes reinforced these themes. The gamified prototype showed that playful competition can encourage participation, though motivation styles vary. The voice diary highlighted the desire for conversational acknowledgment, while the coaching session demonstrated that a balance of praise and critique sustains motivation.

Overall, our findings shifted the project focus toward creating accessible, confidence-building, highly-personalized vocal experiences that lower barriers to entry and treat singing as an emotionally sensitive practice. These insights now inform our ongoing design process and guided

our "how might we" questions around using voice-AI and design to support music pedagogy, rather than replace human creativity and expressive in-person instruction.

5. POVs & Experience Prototypes

a. Your final 1–3 POV statements

POV 1: Ryan K.

We met Ryan, a Stanford alumni budget analyst for Santa Clara County and intermediate jazz pianist who has taken beginner vocal lessons and performed for Stanford's children's theater club, Flying Tree house. **We were surprised to notice** that despite bringing joy and happiness to others through music being one of his primary values and core passions, he doesn't play a lot of music. **We wonder if this means** he would play more music and extend his musical outreach further if the right support and resources made musical expression and performance more accessible to busy professionals. **It would be game changing** to give Ryan, and people like him, the ability to share their music and values easily with a broader audience.

POV 2: Daniel

We met Daniel, a Berkeley AI computer science student who uses voice technology daily for music-related tasks. **We were surprised to notice** that despite relying heavily on voice for commands, he does not find voice AI to feel like a true conversation, especially for music pedagogy. **We wonder if this means** that current voice technology fails to capture the emotional nuance and conversational depth needed for music instruction. **It would be game changing** to make voice-based AI interactions feel more natural and human so voice-AI becomes a viable medium for music pedagogy.

POV 3: Fernando

We met Fernando, a 23-year-old Columbia alum working at a Palo Alto startup who has never played an instrument but dreams of learning piano or violin. **We were surprised to notice** that his biggest barrier to music isn't desire, it's activation energy: his free time comes in short bursts, he has no instrument, and lessons are expensive. **We wonder if this means** he would pursue music if logistical friction, cost, and early beginner discouragement were lesser barriers than they currently are for novice musicians. **It would be game changing** to make first contact with a real instrument and instruction fast, affordable, and motivating so people like Fernando can finally break through beginner-musician-barriers.

b. A sampling of the HMWs that stemmed from each POV**From POV 1 (Ryan)**

- How might we expand the outreach of the perspectives and thoughts of passionate music/vocalist influencers?
- How might we capture life's most beautiful moments through singing?
- **How might we encourage singers to perform with more confidence and passion?** (Top HMW #1)

From POV 2 (Daniel)

- **How might we design voice AI for human connection, reflection, and creativity, rather than just productivity?** (Top HMW #2)
- How might we make silence or slowness part of the voice AI experience as it is in music pedagogy?
- How might we encourage deeper, more sustained pedagogical/artistic engagement with voice AI?

From POV 3 (Fernando)

- How might we bring an instructor and instrument to his dorm/apartment or office on demand?
 - How might we make a hybrid coach (real instrument + smart guidance) feel exciting, not “ed-tech”?
 - **How might we shorten the “I suck for months” phase with fast, meaningful, and audible wins?** (Top HMW #3)
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c. Top 3 solutions from brainstorming

1. **Gamified Acapella App:** a low-stakes, competitive practice environment that rewards participation with leaderboards, points, and badges.
 2. **Artistic Voice Diary:** a voice-based reflective diary for artists that prompts users to speak their thoughts aloud, enabling emotional grounding and self-reflection.
 3. **Collaborative Peer-to-Peer A-Capella Learning:** a mixed positive-and-critical feedback environment for learning music socially and interactively.
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d. Brief description of each experience prototype

1 Prototype 1: Gamified Singing App (Solution #1)

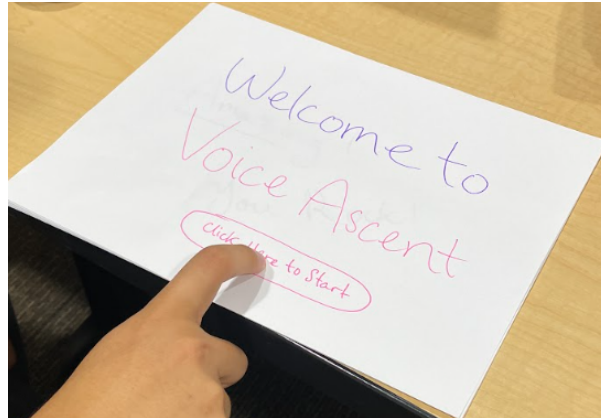


Figure 2: Start screen of our gamified singing app experience prototype.

i. The assumption being tested

Main assumption: Introducing low-stakes competition will motivate Acapella singers to practice singing more.

Supporting beliefs:

- People are naturally competitive.
- Cumulative reward-based systems reduce pressure.
- Gamification can increase participation without losing pedagogical value.

What we're testing: Whether virtual motivational gains can increase musical participation and, overall, overcome a lack of in-person instructional depth.

ii. Key aspects of the prototype setup

- **Participants:** Two Stanford students (sophomore + senior), non-vocalists who enjoy singing casually.
- **Method:** Paper prototype with a landing page, song selection, motivational quotes, leaderboard, and points.
- **Process:** Users selected from three simple songs, sang short excerpts, then saw updated points or ranks.
- **Observation focus:** willingness to sing again, preference for points vs. leaderboard, desire for supplementary instruction.

iii. What worked/didn't work, implications

What worked:

- Non-vocalists sang with confidence and were motivated to sing multiple times.
- Leaderboard strongly motivated one participant while another enjoyed the point system a great deal.
- Simple, familiar songs boosted confidence.

What didn't work:

- One participant preferred critique instead of competition.
- Reward appeal varied — gamified motivation is not universal.

Implications:

- Gamification is powerful but not universally motivating.
- Optional modes (leaderboard vs. instructional feedback) may be necessary.
- Gamified singing is more fun and “addicting” to casual users than expected.

2 Prototype 2: Voice Diary (Solution #2)

i. The assumption being tested

Main assumption: Artists with limited voice-tech experience can still find voice-based reflection meaningful, expressive, and emotionally grounding.

Supporting beliefs:

- Speaking aloud can feel cathartic.
- A judgment-free voice space encourages deeper thought.

What we're testing: Whether voice AI can support a sustained reflective, artistic, and engaging space rather than just providing purely functional commands.

ii. Key aspects of the prototype setup

- **Setup:** Quiet space; facilitator acting as art-focused voice-AI; prompts on index cards; 10–15 min sessions.
- **Method:** Participant spoke aloud about their day, using guided reflection prompts and intentional pauses.
- **Facilitator role:** Prompt only; no judgment; optional summaries.
- **Observations:** Emotional engagement, hesitations, depth of reflection, comfort speaking aloud.

iii. What worked/didn't work, implications

What worked:

- Speaking aloud felt grounding and cathartic.
- Prompts guided reflection effectively.
- Silence intervals deepened introspection.

What didn't work:

- Lacked conversational back-and-forth; felt less human.
- Users were unsure whether they would continue frequent use.

Implications:

- "Human-feeling micro-acknowledgments" are needed in voice AI.
- Younger Acapella users may adopt this more easily.
- Must consider accessibility for accents and speech variance.

Prototype 3: Collaborative Peer-to-Peer A-Capella Learning (Solution #3)

i. The assumption being tested

Main assumption: A balance of criticism and compliments motivate A-capella singers more than only positive or only negative feedback.

Supporting beliefs:

- Encouragement helps boost perseverance.
- Too much criticism is demoralizing.

What we're testing: Whether the right ratio of positive-to-negative feedback boosts motivation and musical improvement.

ii. Key aspects of the prototype setup

- **Participants:** Experienced singer (teacher) + novice singer (student).
- **Method:** 30-minute singing lesson with alternating compliments and criticism.
- **Instruction:** Start balanced, then tilt slightly toward criticism during song practice.
- **Observation focus:** body language, perseverance, emotional response, motivation levels.

iii. What worked/didn't work, implications

What worked:

- Mixture of criticism and compliments was strongly preferred.
- Student felt challenged and motivated.
- One-on-one environment promoted perseverance.

What didn't work:

- Overly harsh criticism briefly reduced confidence.
- The lack of personalization/adaptation to each learner's sensitivity led to a worse overall user experience.

Implications:

- Balanced feedback > purely positive feedback.
- Feedback systems may need personalization based on user tolerance.
- Potential to adapt for users with different cultural or speech backgrounds.

6. Design Evolution

a. Final Solution

Our final solution is an engaging, personalized pedagogical vocal tool for A Cappella singers designed to accelerate artistic growth and help users overcome the progress plateau. The tool enables singers to carry out a full, safe practice regimen with warm-ups, personalized song selection, interactive live feedback sessions with an AI-emulated vocal coach, targeted daily drills, and consistent progress tracking.

The two major problems addressed by our solution are the affordability and accessibility of high-quality vocal instruction.

1. **Affordability.** In-person lessons can cost anywhere from a few hundred to several thousand dollars per week. While our app is not intended to replace professional instruction, it offers an excellent free alternative for A Cappella singers seeking tailored feedback necessary to improve technique and performance skills.
2. **Accessibility.** The student-to-teacher ratio in vocal instruction is extremely poor. The number of talented A Cappella singers far outweighs the number of qualified vocal coaches. As a result, singers often wait long periods between lessons—or struggle to begin lessons at all. Advanced singers require consistent, personalized feedback to meaningfully improve. Our app ensures that A Cappella singers have access to conservatory-level feedback at any time. While in-person instruction is valuable for identifying subtle issues, our app provides continuous monitoring during practice to instantly correct small physical or vocal quality mistakes, helping users make better use of their time in a more consistent and efficient manner than traditional instruction alone can offer.

Tasks

1. **Simple Task: Complete a vocal warm-up session.** Users must complete a 30-second scale warm-up followed by a 30-second lip-burble exercise. To execute the warm-up, they simply follow the visual cues on screen and match the indicated tone with their voice. *Justification:* Warm-ups are essential to any vocal practice routine. For our purposes, a quick warm-up prevents strain or injury, increases user comfort, and leads to higher-quality feedback (greater empowerment). Even advanced singers—such as our target A Cappella user base—must warm up to protect their instrument.
2. **Moderate Task: Choose a song and receive live practice-session feedback.** Users can search for and select a song from the catalog that best suits their interests. After selecting a song, they sing through a one-minute and fifteen-second practice session and receive live feedback from the AI voice-coach avatar. To fully complete this task, the user must finish the session and view their practice summary and performance statistics. *Justification:* Song choice is the centerpiece of any practice session. A Cappella singers benefit most from practicing music they enjoy and feel comfortable with, enabling them to attempt more challenging techniques. Live feedback instantly corrects minor mistakes that may hinder artistic growth or cause injury—mirroring the main advantage of in-person coaching. Because singers cannot be expected to remember every detail from their session, the summary report offers both a high-level and in-depth overview of strengths and weaknesses to guide future practice.
3. **Complex Task: Generate daily drills from mistakes, practice them, and monitor progress.** Users receive automatically generated daily drills that target their most significant weaknesses. They then practice these drills and observe how their progress metrics update over time. *Justification:* To make the process of addressing mistakes more seamless, the daily drill system analyzes performance and produces personalized exercises tailored to A Cappella singers' needs. Without this feature, users would have to identify their weaknesses and design their own practice routines—a burden that may discourage consistent practice. By providing structured, targeted exercises and visible progress tracking, the app encourages sustained improvement and empowers users to recognize how far they have come.

Values in Design

1. **Transparency.** Our final prototype prioritizes transparency by providing real-time, actionable visual feedback. This includes pitch tracking (via a sharp/flat meter), rhythm accuracy (through a metronome animation), and clear AI-generated coaching cues that help users understand their performance quality in the moment. After each session, users receive comprehensive summaries identifying flat or sharp notes, breath support quality, articulation clarity, and rhythmic consistency. Scoring rationales are conveyed through time-aligned graphs, bullet-point explanations, and aggregated statistics. Longitudinal

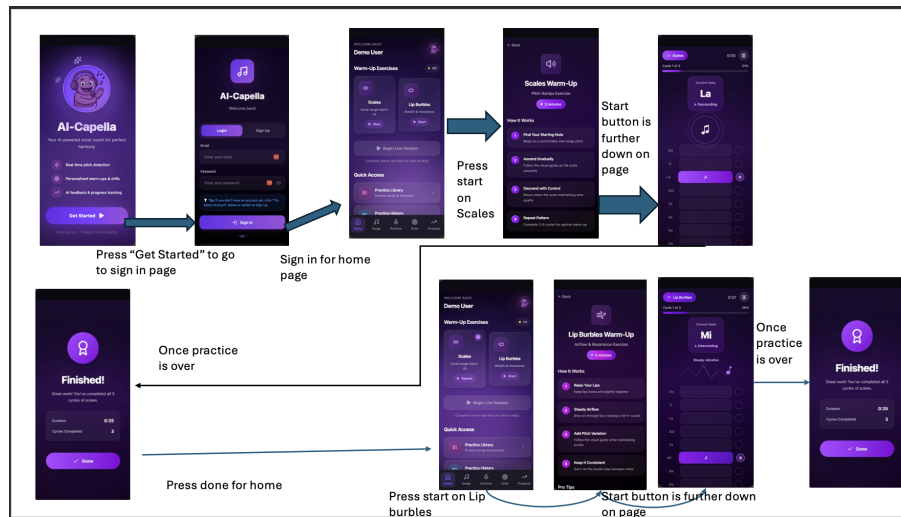


Figure 3: Simple Taskflow

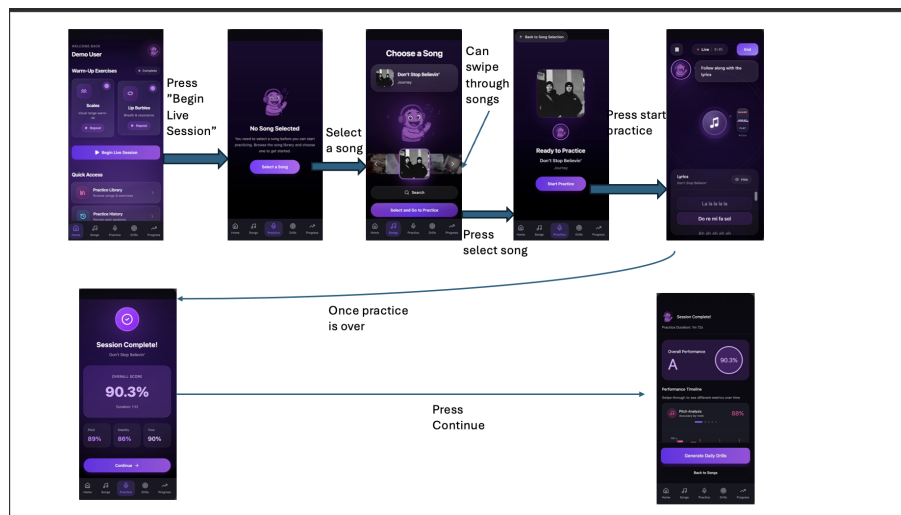


Figure 4: Medium Taskflow

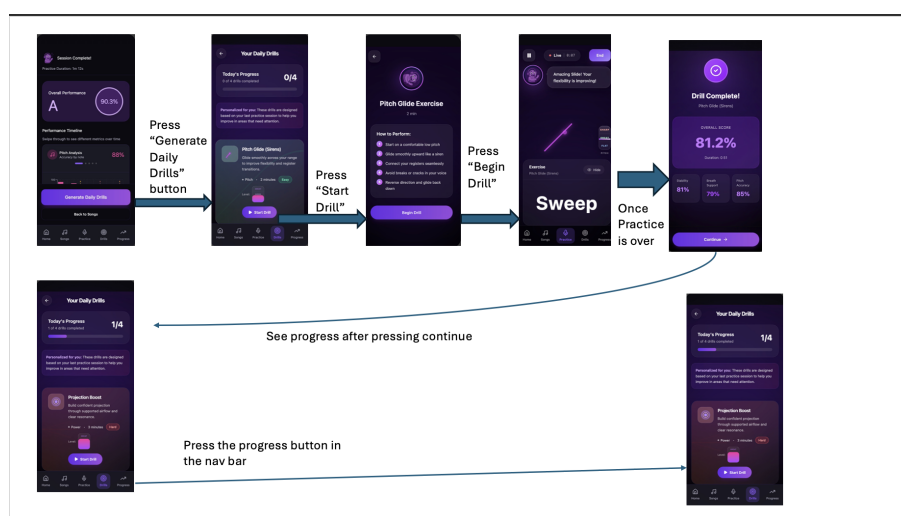


Figure 5: Complex Taskflow

progress is visualized on a dedicated progress page, enabling singers to clearly interpret their growth over time.

2. **Empowerment.** In addition to making feedback transparent, our design emphasizes empowerment through constructive, supportive real-time guidance. The system intentionally avoids discouraging or overly critical phrasing, instead focusing on specific, actionable insights. Progress graphs and metrics reinforce improvement, helping A Cappella singers recognize their development and stay motivated. By framing feedback in an encouraging and confidence-building manner, users feel more capable, less intimidated, and better prepared to perform at future A Cappella events.
3. **Fun.** To ensure sustained engagement, we incorporated playful interactive elements such as the friendly AI-Capella Genie avatar, designed to create a welcoming, low-pressure atmosphere. Custom animations and visually dynamic screens provide an enjoyable complement to instructional content. The song library includes a wide variety of trending and genre-diverse tracks, helping practice sessions feel fresh, exciting, and creatively inspiring rather than repetitive or intimidating.

Value Tensions

1. **Transparency vs. Fun.** Detailed performance analytics can occasionally conflict with the lighthearted, animated design elements intended to make practice sessions enjoyable.
2. **Transparency vs. Empowerment.** While transparent performance data is invaluable, presenting raw critical feedback without care may unintentionally diminish user confidence or discourage continued practice.
3. **Fun vs. Empowerment.** Although animations and playful visuals boost engagement, relying too heavily on them can reduce the perceived credibility or seriousness of the coaching experience.

How We Addressed These Value Tensions

1. **Addressing Transparency vs. Fun.** To balance clarity and enjoyment, users first encounter high-level celebratory feedback immediately following a practice session. More detailed statistics are accessible on the subsequent “Generate Daily Drills” screen. This sequencing maintains the uplifting tone of the experience while still ensuring transparency. We also established feedback phrasing constraints to ensure comments remain constructive and supportive—for example, reframing “you were flat” as “try raising your pitch slightly next time.”
2. **Addressing Transparency vs. Empowerment.** To preserve user confidence while delivering honest feedback, we emphasized trend-based visualizations (e.g., progress from 70% to 75%) rather than reductive labels or grades. Complementary bullet points and contextual explanations help users interpret their improvements clearly. This approach

provides full visibility into performance without framing shortcomings in a discouraging or demotivating way.

3. **Addressing Fun vs. Empowerment.** Animations and playful elements were intentionally limited to supportive, non-intrusive moments and were carefully positioned so they never interfere with instructions, lyrics, or pitch tracking. This maintains a professional and credible learning environment while still feeling friendly and engaging. Additionally, many fun elements are earnable through achievements on the progress screen, ensuring that enjoyment reinforces, rather than detracts from, meaningful practice and artistic development.

7. Final Prototype Implementation

a. Tools used; pros and cons

Our final prototype for AI-Capella consists of two layers:

- A high-fidelity interaction prototype built in **Figma Make** to rapidly explore and communicate the full user experience.
- A fully functional **React Native** app (with Expo) backed by **Supabase**, integrated with the **Claude API** for progress generation and **ElevenLabs** for voiced feedback.

Together, these allowed us to test both the experience design and the technical feasibility of a real-time AI vocal coach.

Figma Make (AI-assisted UI prototyping)

We used Figma Make, an AI-powered design tool, to quickly generate and refine the visual and interaction design of AI-Capella.

Pros:

- Allowed us to generate entire screens and flows from text prompts, dramatically speeding up early exploration.
- Supported complex navigation and screen-to-screen flows, so participants experienced something that felt like a real app during usability tests.
- Made it easy to maintain visual consistency (colors, typography, spacing) across many screens.

Cons:

- No access to real microphone or audio processing; all “live feedback” had to be simulated or later reproduced in the coded app.
- AI-generated layouts and components often needed manual cleanup to match our brand and accessibility standards.

- Designs could not be exported directly into clean React Native code, so we still needed a translation step into implementation.

React Native with Expo (cross-platform mobile app)

Our production-facing prototype was implemented in React Native and run via Expo, targeting iOS, Android, and web.

Pros:

- Single codebase for multiple platforms.
- Expo's dev tooling allowed rapid iteration on flows and UI.
- Direct access to microphone and audio playback via Expo's modules, enabling us to build real recording and feedback loops.

Cons:

- Advanced low-latency audio analysis (e.g., robust real-time pitch tracking) is difficult to implement purely in JavaScript, and would likely require native modules.
- Managing audio state was non-trivial and introduced edge cases that were more engineering-heavy than design-focused.

Backend

- **Supabase:** For authentication, user accounts, and storing structured data such as songs, lyrics, session history, and progress metrics.
- **Claude API:** For generating written practice summaries and personalized coaching suggestions based on session metadata and simple metrics.
- **ElevenLabs:** For generating natural-sounding audio snippets of the AI-Capella Genie's spoken feedback, so users hear their coach as a consistent, warm voice rather than just reading text.

b. Wizard of Oz techniques used

While our final prototype includes a real backend and AI integrations, several intelligent behaviors are still implemented using Wizard of Oz techniques. These approaches simulate deeper vocal analysis and personalization so that the product feels cohesive and complete, even though some underlying systems are not yet fully developed.

Simulated Multi-Dimensional Vocal Analysis

The interface suggests that AI-Capella evaluates dimensions such as pitch accuracy, rhythm consistency, breath support, and articulation. In reality, our only real-time signal is loudness, combined with basic session metadata (song, duration, completion). We map these simple inputs

to higher-level categories. From the user's perspective, the coach appears to understand many aspects of their performance, but behind the scenes, these are approximations.

Scripted Live Feedback from the Genie

During warm-ups and song practice, the AI-Capella Genie appears to react to what the user just sang with comments. While the app does use some backend functionality to decide what feedback to give, some comments are timed and pre-scripted for specific points in the session.

Semi-Personalized Daily Drills

After each session, the app claims to analyze the user's mistakes and generate daily drills. Under the hood, we store session history in Supabase and pass simplified information (e.g., length of session, loudness patterns, selected goals) to Claude. Claude then selects and describes drills from a predefined pool of exercise types that we designed.

Why These Wizard of Oz Techniques Were Used

These Wizard of Oz techniques allowed us to present a polished, believable AI coach while focusing this class project on experience design and core workflows. By simulating deeper ML analysis and personalization, we could:

- Test whether singers actually value this style and tone of feedback.
- Experiment with the right level of detail in feedback (high-level vs. highly granular).
- Gather meaningful reactions from users about the concept of an always-available AI vocal coach and keep implementation manageable.

c. Hard-coded techniques used

To support realistic demonstrations, several components of the app are hard-coded. These elements give users a clear sense of how AI-Capella will behave in a production setting.

Hard-coded Song Library

The app includes a curated, fixed set of songs with manually authored metadata, difficulty tags, and recommended practice sections. Lyrics are entered by hand rather than imported from external music services. Users cannot yet upload their own songs or import arbitrary tracks. All the song data is stored in a Supabase database, not hard-coded locally, so we could push more songs without updating the app, but it is still a limited set right now.

Fixed Thresholds and Predefined Genie Lines

Loudness thresholds for “too quiet,” “okay,” and “strong” projection are fixed values determined through trial and error and do not adapt dynamically to the user’s environment or voice type. We then map these thresholds to more nuanced labels (e.g., “try using more breath support”) through static rules. Similarly, although ElevenLabs can generate arbitrary speech, we pre-generated a library of common Genie lines (praise, nudges, and instructions). At runtime, the app selects appropriate lines from this library rather than synthesizing entirely new sentences for every interaction, which keeps the character’s voice consistent, feedback quick, and our costs minimal, since ElevenLabs can be very expensive to generate speech.

Why These Elements Were Hard-Coded

Hard-coding these components allowed us to ship a stable prototype with limited development time while still demonstrating the full end-to-end experience: selecting a song, practicing with an AI coach, receiving feedback, and viewing progress. This approach ensured that:

- Key flows worked reliably during usability testing and live demos.
- We could focus on interaction quality.
- In future iterations, we can gradually replace hard-coded content and thresholds with more robust, data-driven and model-driven implementations.

d. AI tools used with justification

Figma Make

We used Figma Make’s generative features to:

- Produce initial screen layouts from textual descriptions of key flows (warm-ups, practice sessions, progress pages).
- Explore multiple design directions quickly and then manually refine the best ones.
- Experiment with interface variations for displaying feedback (e.g., graphs vs. badges vs. textual lists).

Justification: Using Figma Make allowed us to move from concept to high-fidelity screens extremely quickly, so we could spend more time testing flows with users rather than hand-drawing every screen from scratch.

Cursor and Claude Code

We used AI-assisted coding primarily to:

- Generate boilerplate React Native components and Supabase SDK integration.

- Refactor and clean up the old React web code exported from Figma Make.
- Speed up integration of external APIs (Claude, ElevenLabs).

Justification: Using Cursor and Claude Code allowed us to move much faster than we could have with manual coding alone. These tools handled repetitive boilerplate (navigation, API wrappers) and suggested refactors for our existing React web code from Figma Make, so we could focus our time on higher-level interaction design, data modeling, and debugging tricky audio flows.

8. Reflection & Next Steps

Key Learnings About the Design Thinking Process

One of our biggest learnings this quarter was how much our initial assumptions about singing, feedback, and AI had to be softened once we actually sat down with users. Going into needfinding, it was tempting to project our own experiences as student musicians onto our participants and ask leading questions like, “Wouldn’t it be nice if . . .” Instead, focusing on open-ended prompts like “Tell us about a time you felt discouraged while singing” surfaced richer, more surprising stories than any of us anticipated.

We heard about people nearly quitting singing after one harsh critique, about singers who loved music but avoided practice because they were afraid of finding out how bad they really are, and about users who felt oddly more comfortable being judged by an AI than by another human.

We also experienced firsthand how critical it is to iterate across fidelity levels instead of jumping straight into code. Our early paper sketches and Figma Make prototypes gave us a low-stakes way to test flows like “start a warm-up → sing → see feedback” before we worried about latency, audio APIs, or database schemas. Many interaction details came directly from watching people struggle with our early designs.

Finally, we learned to respect the tension between what is technically cool and what is usable. There were several times we got excited about ambitious features (multi-part harmony analysis, more complex metrics) and then realized they would overwhelm users in vulnerable moments. The design process forced us to repeatedly ask: “Is this making the singer feel more confident and supported, or more judged and confused?”

Key Learnings About the Voice AI Studio Theme

Our studio theme, Voice AI, pushed us to think beyond voice as just another input/output modality and toward voice as a medium with its own emotional and pedagogical properties. Early on, it was easy to treat Voice AI like “text chat, but spoken.” Through the quarter, we learned the nuances.

First, we saw how strongly users react to the tone and pace of a voice. The exact same message (“You were a bit flat on that note”) can feel encouraging or brutal depending on how it is

delivered. Several participants explicitly said they would tolerate much harsher criticism from a synthetic coach than from a human teacher.

Second, we learned that Voice AI for pedagogy is fundamentally different from Voice AI for commands. Many existing assistants optimize for speed: you say something, they respond instantly, and then the interaction is over. Our use case required something closer to a conversation across time. Singers did not want the coach to talk over them or interrupt their musical focus. Designing around these constraints changed how we structured feedback into shorter, well-timed comments.

Third, we discovered that voice interfaces carry risks. Latency, mispronunciations, or a mismatched accent can break trust quickly. The Voice AI theme forced us to confront these trade-offs, especially around latency.

Overall, the studio theme helped us see Voice AI as more than a technology trend.

Key Learnings About AI-Capella

Through building AI-Capella, we learned that the value of our tool is not just in correcting notes, but in shaping the emotional context of practice. The interviews made it clear that many singers already know they are imperfect; what they lack is a space where imperfection feels safe enough to work with.

This insight influenced almost every design decision:

- We framed feedback as coaching, not judging.
- We made sure positive reinforcement appeared before detailed metrics.
- We gave the Genie a playful visual identity to soften otherwise blunt critiques.

We also learned how unforgiving real-time systems can be. Once we moved from Figma Make into React Native and Supabase, we had to manage audio recording, API calls to Claude and ElevenLabs, and UI transitions without breaking the illusion of a responsive coach. This process taught us a lot about scoping: which parts of the “perfect” AI coach needed to be real right now (e.g., storing real progress, playing back real TTS), and which could be approximated via Wizard of Oz techniques (e.g., detailed pitch analysis).

Lastly, AI-Capella taught us that there is no single thing simply called “AI.” The system we built is actually a stack of simpler pieces: loudness detection, drill templates, structured prompts to Claude, and a fixed library of Genie lines. This modular approach gave us confidence that we can gradually swap in more advanced models in the future.

Next Steps

The next steps for our project could be:

Richer Audio Analysis and Vocal Metrics

Our first priority would be building a more robust audio analysis pipeline. This would include pitch tracking, rhythm alignment, and basic timbre features so that feedback feels more specific and musically grounded. These metrics could then feed into Claude in a structured way, allowing the Genie to make comments that are both emotionally sensitive and technically accurate.

Personalized Coaching Styles

Different singers want different coaching styles: some prefer gentle encouragement, others want blunt critique. With more time, we would add a way for users to choose or gradually learn their preferred feedback style. The Genie's tone, frequency of interruptions, and drill selection could all adapt based on these preferences and how the user responds over time.

Expanded Content Library

Currently, our song library is curated and limited. In the future, we would explore ways to safely scale. We would also like to support multiple genres and difficulty levels, as well as short skill-focused snippets rather than only full songs.

Body and Technique Feedback via Computer Vision

Several viewers at the expo reminded us that a singer's voice is more than just their vocal cords; the entire body has to act like an instrument. We want to experiment with an optional camera-based mode that provides gentle feedback on posture, neck tension, and breath mechanics.

Social and Ensemble Features

Many a cappella singers practice in groups, not just alone. In the long term, we would like to explore features that support ensemble rehearsals with shared warm-ups, group pitch-center detection, or post-rehearsal summaries that can be sent to the whole group. We also imagine a "safe sharing" mode where users can share selected practice clips or progress summaries with trusted peers, voice teachers, or directors to get human feedback on top of the AI coach.